Transformations of Conics and the Erlangen View of Geometry

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§1.1 The organizing question

This chapter studies conic sections through transformations. The guiding question is not only how an ellipse, parabola, or hyperbola is described by an equation, but what kind of geometry regards two such objects as the same.

The hierarchy is:

$$\text{Euclidean} \subset \text{similarity} \subset \text{affine} \subset \text{projective} \subset \text{conformal/inversive phenomena.}$$

Each enlargement forgets some invariants and preserves others. This is the spirit of Klein's Erlangen program: a geometry is the study of properties invariant under a chosen transformation group.

§1.2 Affine transformations

Definition 1.2.1. An affine transformation of $\mathbb{R}^n$ is a map

$$T(x) = Ax + b,$$

where A is a linear map and $b \in \mathbb{R}^n$. If $A \in GL(n, \mathbb{R})$, then T is an affine automorphism.

Affine transformations preserve straight lines, parallelism, ratios along a line, convexity, and barycentric combinations. They do not preserve lengths, angles, or circles.

Example 1.2.2

An ellipse can be sent to a circle by an affine transformation. If

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

then the affine map

$$(x, y) \mapsto (x/a, y/b)$$

sends it to the unit circle.

§1.3 Conics in matrix form

A plane conic may be written as

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

In homogeneous coordinates $[x : y : z]$, this becomes

$$Ax^2 + Bxy + Cy^2 + Dxz + Eyz + Fz^2 = 0.$$

Equivalently,

$$X^T Q X = 0, \qquad X = (x, y, z)^T,$$

where Q is a symmetric 3×3 matrix.

Under a projective transformation $X \mapsto MX$, the conic transforms by

$$Q \mapsto M^{-T} Q M^{-1}.$$

Thus projective classification of conics becomes a question about symmetric bilinear forms up to change of coordinates.

A concrete matrix is the best antidote to the vague sentence that all nondegenerate conics are projectively equivalent. The parabola

$$C_1 : \quad y = x^2$$

has homogeneous equation

$$X^2 - YZ = 0,$$

so it is represented by

$$A_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} \\ 0 & -\frac{1}{2} & 0 \end{pmatrix}.$$

The unit circle

$$C_2 : \quad X^2 + Y^2 = Z^2$$

is represented by

$$A_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Now take

$$M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{pmatrix}.$$

A direct multiplication gives

$$M^T A_1 M = A_2.$$

Indeed, if $U = (u, v, w)^T$ and $X = MU = (u, v + w, -v + w)^T$, then

$$X^T A_1 X = u^2 - (v + w)(-v + w) = u^2 + v^2 - w^2 = U^T A_2 U.$$

Thus the projective change of coordinates $U = M^{-1} X$ sends the parabola equation to the circle equation.

This is not an affine transformation. The inverse matrix is

$$M^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

The unique real point at infinity of the parabola,

$$[0 : 1 : 0],$$

is sent to

$$M^{-1}[0 : 1 : 0] = [0 : 1 : 1],$$

a finite real point on the circle. Meanwhile the usual circular points at infinity of the real circle are the complex points

$$[1 : i : 0], \qquad [1 : -i : 0].$$

So the projective transformation has moved the affine line at infinity itself; this is exactly why a parabola and a circle can be the same projective conic while remaining different in affine geometry. The difference between them is absolute for affine geometry, but only a coordinate choice for projective geometry.

§1.4 Projective geometry

Projective geometry adds points at infinity. In the real projective plane,

$$\mathbb{P}^2(\mathbb{R}) = \{[x : y : z] \mid (x, y, z) \neq 0\}/\sim,$$

where

$$[x : y : z] = [\lambda x : \lambda y : \lambda z], \qquad \lambda \neq 0.$$

The affine plane is the chart $z \neq 0$, and the line at infinity is $z = 0$.

Definition 1.4.1. A projective transformation of $\mathbb{P}^2$ is induced by an invertible matrix

$$M \in GL(3, \mathbb{R}),$$

acting by

$$[X] \mapsto [MX].$$

Scalar multiples of M induce the same projective transformation. Thus the transformation group is

$$PGL(3, \mathbb{R}).$$

Projective transformations preserve incidence and cross-ratio, but not parallelism, distance, or angle. In projective geometry, all nondegenerate conics over an algebraically closed field are projectively equivalent.

§1.5 Cross-ratio

For four collinear points A, B, C, D, the cross-ratio is denoted

$$(A, B; C, D).$$

In an affine coordinate it has the form

$$(a, b; c, d) = \frac{(c-a)(d-b)}{(c-b)(d-a)}.$$

Theorem 1.5.1

Projective transformations preserve cross-ratio.

Remark 1.5.2 — The cross-ratio is the prototype of a projective invariant. Once one understands it, projective geometry stops looking like distorted Euclidean geometry and becomes an invariant theory of incidence.

§1.6 Projective duality and conics

Projective duality interchanges points and lines. A nondegenerate conic defines a duality: to a point p, one associates its polar line with respect to the conic. In matrix form, if the conic is $X^T Q X = 0$, then the polar line of p is

$$p^T Q X = 0.$$

Remark 1.6.1 — The same quadratic form that defines the conic also turns points into lines. The conic is therefore not just a curve; it is a device for dualizing projective geometry.

§1.7 Inversion

Definition 1.7.1. Inversion in the circle of radius r centered at the origin is the map

$$x \mapsto \frac{r^2}{\|x\|^2} x.$$

In the complex plane, inversion in the unit circle is essentially

$$z \mapsto \frac{1}{\bar{z}}.$$

Inversion sends lines and circles to lines and circles, with the convention that lines are circles through infinity. It is conformal away from the center of inversion: it preserves angles but not distances.

§1.8 Möbius transformations

A Möbius transformation has the form

$$z \mapsto \frac{az+b}{cz+d}, \qquad ad - bc \neq 0.$$

It acts naturally on the Riemann sphere

$$\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}.$$

The group of Möbius transformations is

$$PGL(2, \mathbb{C}).$$

Möbius transformations send generalized circles to generalized circles and preserve cross-ratios. They are simultaneously projective transformations of $\mathbb{P}^1(\mathbb{C})$ and conformal automorphisms of the Riemann sphere.

§1.9 Conic sections

Classically, conics arise as plane sections of a cone. Analytically, they are degree-two plane curves. Projectively, a conic is a homogeneous quadratic equation in $\mathbb{P}^2$.

Geometry	Transformations	Conic classification
Euclidean	Rigid motions	Metric shape matters
Similarity	Rotations, translations, scalings	Eccentricity-like shape remains
Affine	Invertible affine maps	Ellipses are equivalent to circles
Projective	$PGL(3)$	Nondegenerate conics are equivalent over algebraically closed fields

§1.10 The Erlangen program

Klein's Erlangen program says that geometry should be understood through a group of transformations and the invariants under that group. Euclidean geometry studies invariants under Euclidean motions; affine geometry studies invariants under affine transformations; projective geometry studies invariants under projective transformations.

Remark 1.10.1 — This viewpoint explains why the same object can look different in different geometries. A circle is special in Euclidean geometry, but not in affine geometry. Parallel lines are meaningful in affine geometry, but not in projective geometry. Cross-ratio is invisible in elementary Euclidean geometry, but central in projective geometry.

§1.11 A hierarchy of invariants

Transformation group	Preserved	Forgotten
Euclidean	Distance, angle, incidence	Coordinates
Similarity	Angle, ratios of lengths	Absolute scale
Affine	Incidence, parallelism, ratios on a line	Angle and length
Projective	Incidence, cross-ratio	Parallelism and affine ratio
Möbius	Generalized circles, angles, cross-ratio on $\widehat{\mathbb{C}}$	Euclidean size

§1.12 Quadratic forms and matrix language

A conic in the projective plane can be written as

$$X^T Q X = 0,$$

where $X = (x : y : z)^T$ is a homogeneous coordinate vector and Q is a symmetric matrix. Passing to homogeneous coordinates is not cosmetic. It lets ellipses, parabolas, and hyperbolas be treated uniformly and allows points at infinity to participate in the geometry.

Under a projective change of coordinates $X = MX'$, the conic matrix transforms as

$$Q' = M^T Q M$$

or equivalently $Q \mapsto M^{-T} Q M^{-1}$, depending on whether one transforms coordinates or points. The equation changes, but the geometric conic is the same object in new coordinates.

§1.13 Affine classification versus projective classification

In affine geometry, ellipses, parabolas, and hyperbolas are genuinely different because the line at infinity is fixed. The way a conic meets the line at infinity distinguishes the types: an ellipse has no real points at infinity, a parabola is tangent to the line at infinity, and a hyperbola meets it in two real points.

In projective geometry over an algebraically closed field, all nonsingular conics are projectively equivalent. Projective geometry forgets metric shape and focuses on incidence and cross-ratio-type data.

§1.14 Polar lines

For a nonsingular conic $X^T QX = 0$, a point P has a polar line

$$P^T QX = 0.$$

If P lies on the conic, the polar line is the tangent line at P. If P lies outside, the polar encodes the chord of contact of tangents from P.

The pole-polar relation is one of the best examples of why projective conic theory is richer than coordinate classification. It converts points to lines and lines to points in a way controlled by the conic.

§1.15 The structural moral

The transformation-centered view naturally leads to modern geometry:

(i) algebraic geometry studies varieties and morphisms, with projective space as a basic ambient object;

(ii) differential geometry studies invariants under diffeomorphisms or isometries;

(iii) complex geometry studies holomorphic maps;

(iv) representation theory studies symmetries through linear actions;

(v) category theory abstracts the entire pattern by treating structure-preserving maps as primary.

§1.16 How to choose the right geometry

The Erlangen viewpoint becomes useful only when it tells us which tools are allowed. For conic problems, a practical guide is:

Question being asked	Natural geometry	Reason
length, angle, focus, eccentricity	Euclidean/similarity	metric data is essential
midpoints, parallelism, area ratios	affine	affine maps preserve barycentric structure
incidence, tangency, conic equivalence	projective	the line at infinity may move
cross-ratio or four points on a line	projective	cross-ratio is projectively invariant
angle with generalized circles	conformal/inversive	inversion preserves angles and generalized circles

Remark 1.16.1 (A common mistake) — Saying that two conics are projectively equivalent does not mean they are the same in Euclidean or affine geometry. A parabola and a circle may be projectively equivalent, but the projective map can move the line at infinity. Once that happens, affine notions such as parallelism and the distinction between ellipse/parabola/hyperbola need not be preserved.

§1.17 Operational summary

A conic in homogeneous coordinates is

$$X^T AX = 0.$$

Projective changes of coordinates act by congruence on the matrix, and the same bilinear form controls tangents, polar lines, and dual conics. Cross-ratio controls configurations on lines. Inversion belongs to the conformal side because it preserves angles and generalized circles, not lengths.

The central lesson is therefore not merely that there are many transformations. The lesson is to choose the transformation group whose invariants match the question. Once the group is chosen, the correct invariants and the permissible simplifications become clear.